

LAB

Build an AI agent using IBM watsonx Agent Lab

IBM **SkillsBuild**

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Introduction

In this lab, you'll design and deploy a personalized AI agent using watsonx Agent Lab to support fitness, meal planning, and daily health guidance. This drag-and-drop tool enables you to build and validate a real-world healthcare solution without any coding experience.

Software requirements

To complete this lab, you'll need access to an IBM Cloud account with watsonx.ai and Agent Lab enabled. These platforms provide the tools to build and deploy your AI agent. You'll also need a modern web browser such as Chrome, Firefox, or Edge to access the lab environment smoothly.

Basic familiarity with AI concepts or low-code platforms might help you better understand the process.

Objective

After completing this lab, you should be able to:

- Build an AI agent using IBM watsonx.ai

Lab steps

This lab requires you to complete the following procedures:

- Log in to watsonx.ai
- Configure the project for Agent Lab
- Build and configure the Personal Health Advisor agent
- Deploy and validate the agent using visual tools

Estimated duration to complete

30 minutes

Scenario

Background information

A wellness startup called VitaWell provides digital health solutions to its users, including fitness tracking, workout routines, and meal planning. As its customer base grows, the company is exploring the use of a virtual AI agent to deliver personalized, always-available support tailored to each user's goals and health conditions. You've been hired as the lead AI designer to develop this AI-enabled Personal Health Advisor using watsonx Agent Lab.

Challenge

Currently, the team relies on manual content delivery for health advice through blogs and chat scripts. Health coaches are overworked, which leads to delayed response times, generic guidance, and limited user interaction. The company needs a more interactive, responsive, and personalized experience that adapts to each user's unique lifestyle and goals.

Commented [TK1]: Note to IBM: Watsonx.ai is a larger platform with multiple AI capabilities. Agent Lab is one such capability within Watsonx.ai, which allows users to build and deploy agents.

Suggestion: We can update the objective here and in the Rise course to "Build an AI agent using IBM watsonx Agent Lab". This will also ensure consistency with the titles of the Sim and the Lab.

Solution

You'll use watsonx Agent Lab to build a dynamic AI agent powered by the llama-3-3-70B-instruct model. The agent will gather user information and preferences, retrieve contextual data using tools, generate health plans, provide meal suggestions, and curate fitness routines while maintaining a conversational tone and protecting users' private data.

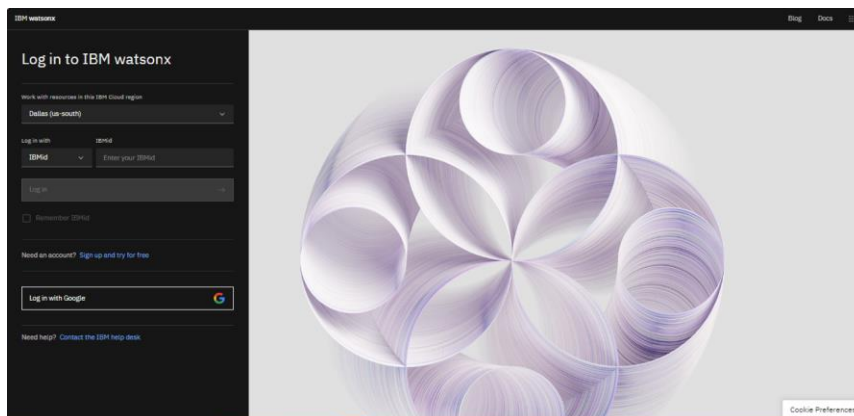
Log in to watsonx.ai

Overview

In this procedure, you'll create or sign in to your IBM Cloud account to access the watsonx.ai environment. IBM Cloud provides the infrastructure and tools needed to build, deploy, and manage AI solutions. You'll develop your AI agent on watsonx.ai, using built-in tools and features that enhance its ability to interpret prompts and respond to people. This setup ensures you have the environment and access needed to build and run the agent powered by the llama-3-3-70B-instruct model.

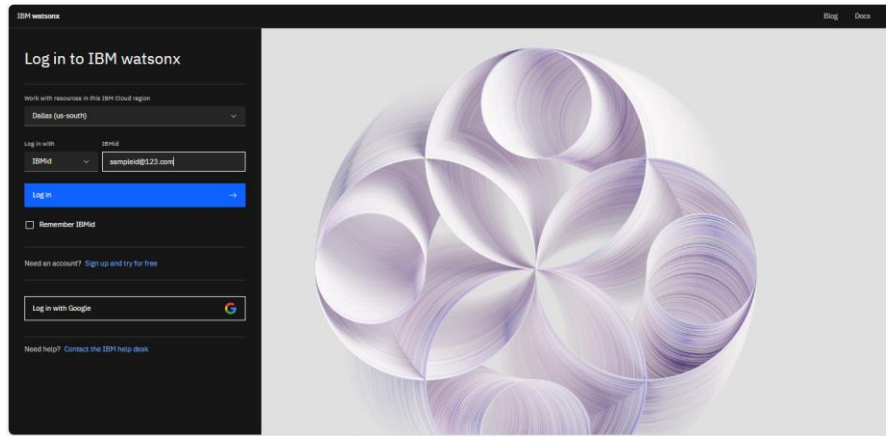
Instructions

1. Go to the [watsonx.ai platform](https://watsonx.ai) and select the **Sign up and try for free** option, as shown in the following screenshot.

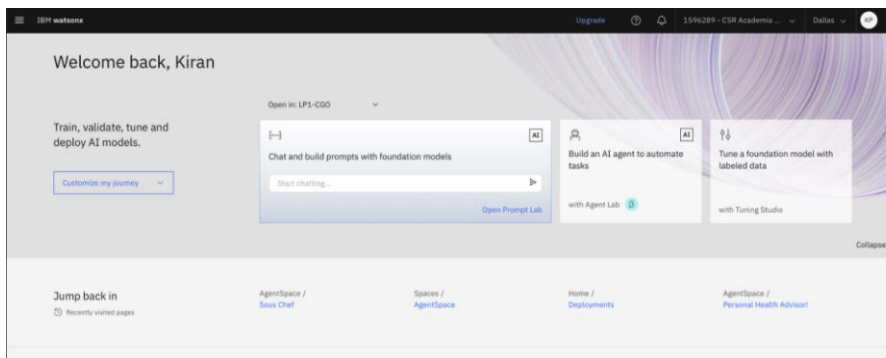


2. Enter your **email** address, create a **password**, and fill in other account details. Then, select **Create account** or **Log in**.

The following screenshot shows the “Log in to IBM watsonx” window.



3. **Result:** After you’ve signed in, the “watsonx.ai” home page is displayed.



Configure the project for the Agent Lab environment

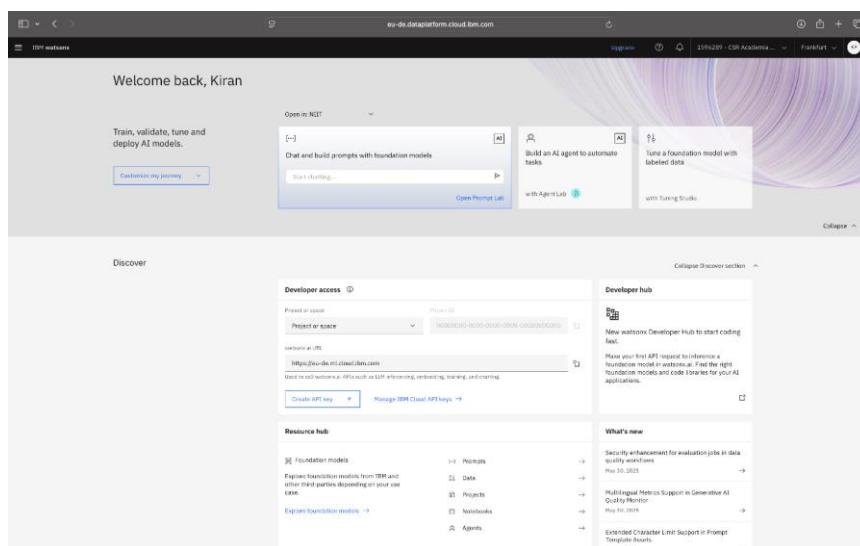
Overview

In this procedure, you'll set up your project environment and choose the runtime that will execute your AI agent in watsonx Agent Lab. This setup is essential to ensure that the agent runs smoothly and has access to the right tools and resources.

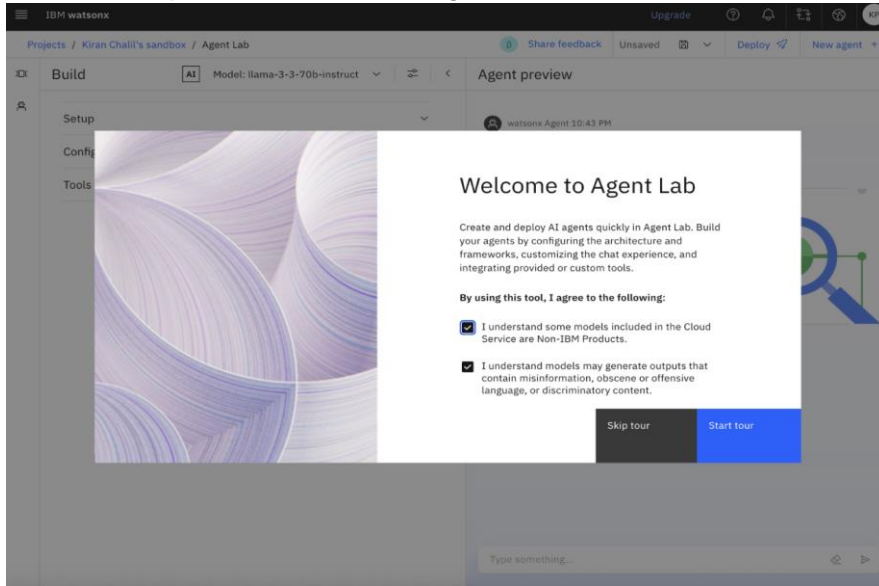
Instructions

1. On the “watsonx.ai” home page, select **Build an AI agent to automate tasks**.

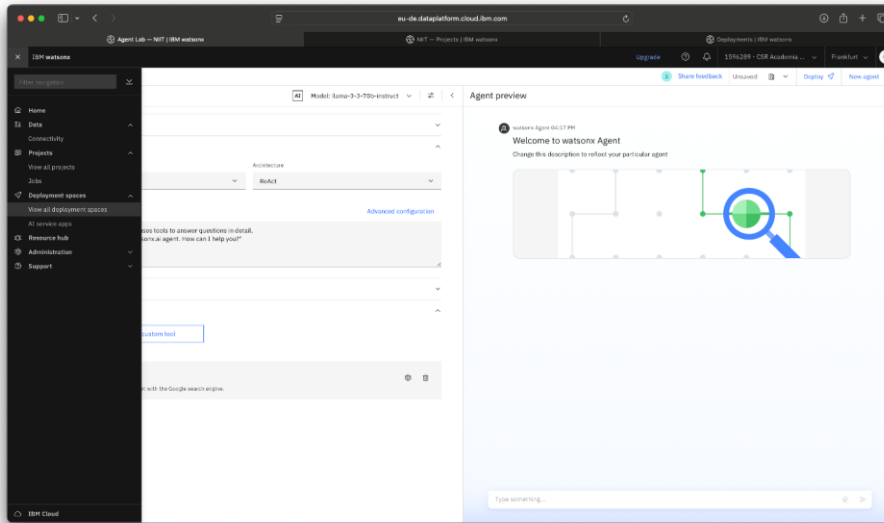
The following screenshot shows the “watsonx.ai” home page.



2. From the Welcome to Agent Lab dialog box, select the recommended checkboxes, and then select **Skip tour** as shown in the following screenshot.



3. From the navigation menu, select **View all deployments spaces** under the **Deployment spaces** section, as shown in the following screenshot.

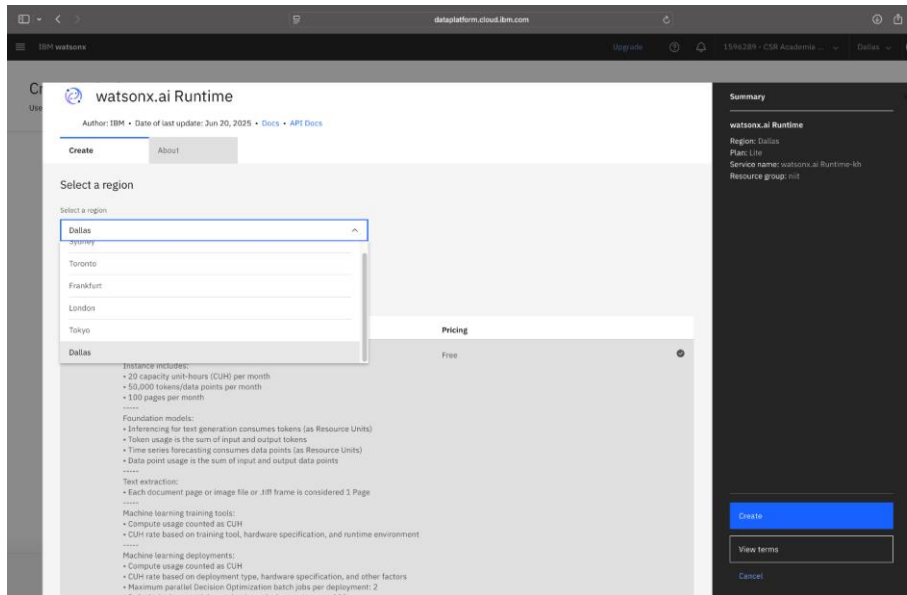


4. In the “Create a deployment space” window, select **Create** to create a space and runtime as shown in the following screenshot.

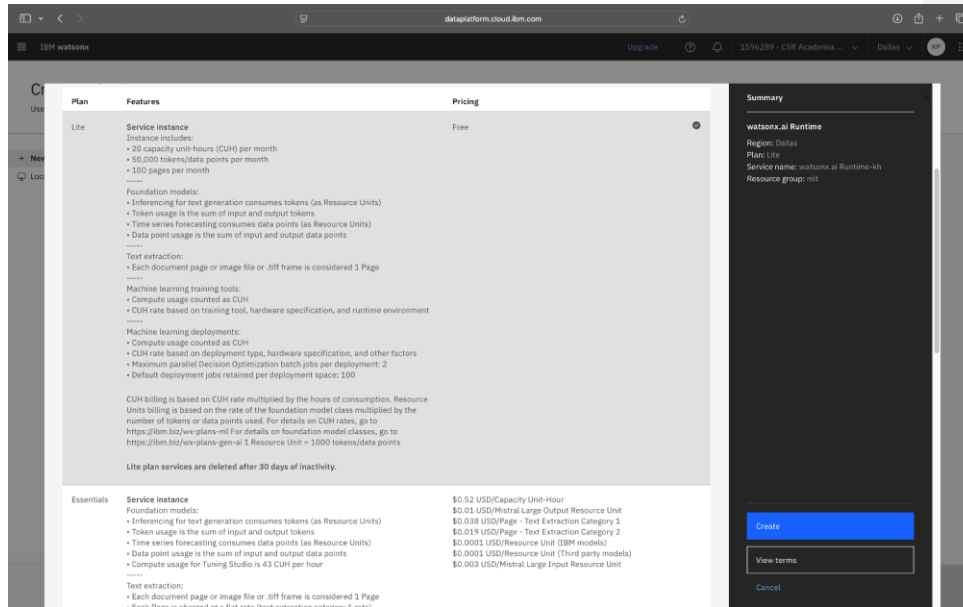
The screenshot shows the 'Create a deployment space' window in the IBM watsonx Agent Lab interface. The window is titled 'Create a deployment space' and has a subtitle 'Use a space to collect assets in one place to create, run, and manage deployments'. On the left side, there is a sidebar with a 'New' button and a 'Local file' option. The main area is titled 'Define details' and contains several form fields: 'Name' (with a placeholder 'Enter a name'), 'Description (Optional)' (with a placeholder 'What's the purpose of this space?'), 'Deployment stage' (a dropdown menu with 'Select or enter a name that describes the purpose of the space'), 'Tag (Optional)' (a dropdown menu with 'Add or create tags'), 'Add tags to make assets easier to find', 'Storage' (a dropdown menu with 'Cloud Object Storage-Ag'), and 'Advanced Settings' (a dropdown menu). A blue 'Create' button is located at the bottom left of the form. At the bottom right of the window, there are 'Cancel' and 'Create' buttons.

5. Select **Dallas** as the region from the drop-down menu.

The following screenshot shows the the setup screen for provisioning the watsonx.ai Runtime service in IBM Cloud and the option to select a region.



- 6 Select **Lite (Free tier)** as the pricing plan, and then select the **Create** option as shown in the following screenshot.



7. After the service is created, refresh the page to return to the deployment creation screen. Type **AgentLABdeployment** in the **Name** field as the name of the deployment space, and then select **Next** to continue.

Create a deployment space

Use a space to collect assets in one place to create, run, and manage deployments

Define details

Name: AgentLABDeployment

Description (Optional): What's the purpose of this space?

Deployment stage: Select or enter a name that describes the purpose of the space

Tags (Optional): Find or create tags

Add tags to make assets easier to find

Storage: Cloud Object Storage-tp

Space will include integration with [Cloud Object Storage](#) for storing space assets.

Watson Machine Learning (Optional): Select or create an instance

Advanced Settings

Cancel Create

8. Select **watsonx.ai Runtime** from the drop-down menu **Select or create an instance**, and then select **Create**.
- The following screenshot shows the “Create a deployment space” window, which includes the field for selecting or creating an instance.

Create a deployment space

Use a space to collect assets in one place to create, run, and manage deployments

Define details

Name
AgentLABdeployment

Description (Optional)
What's the purpose of this space?

Deployment stage
Select or enter a name that describes the purpose of the space

Tag (Optional)
Find or create tags

Add tag to make assets easier to find

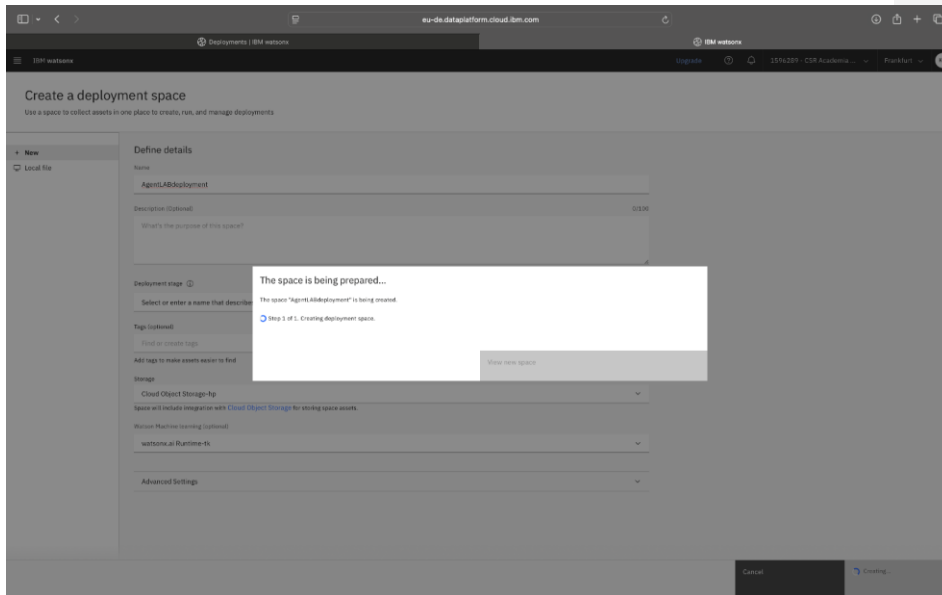
Storage
Cloud Object Storage-ly
Space will include integration with [Cloud Object Storage](#) for storing space assets.

Watson Machine Learning (Optional)
Select or create an instance
Create a new watsonx.ai Runtime service

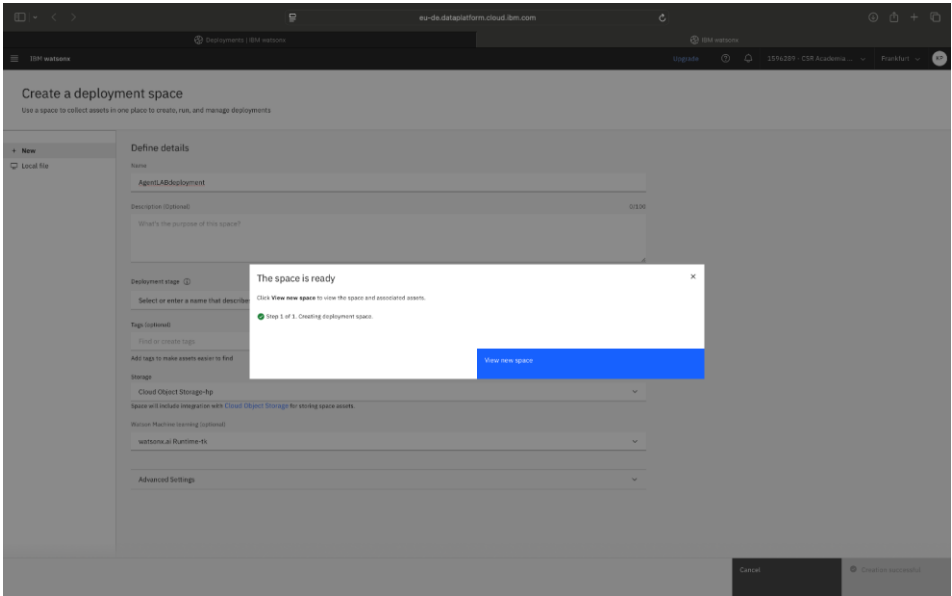
watsonx.ai Runtime-ly

Cancel Create

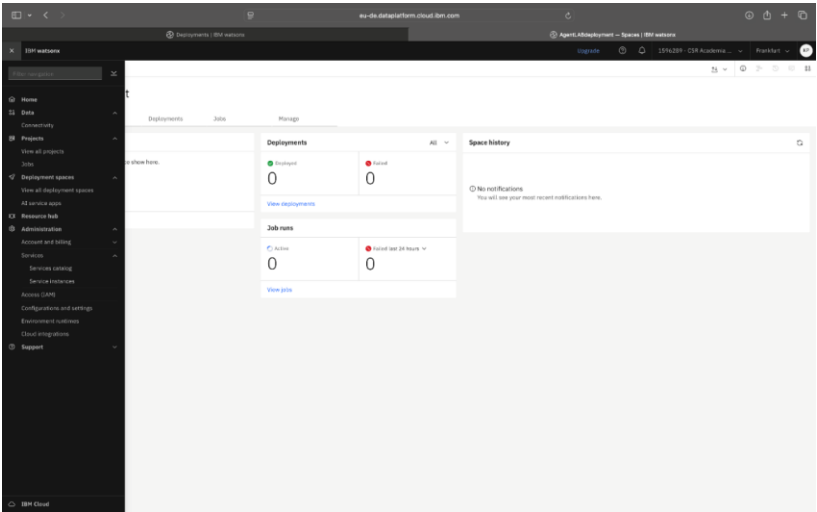
9. **Result:** The deployment space creation is displayed being in progress. Wait for 1–2 minutes for the process to finish.



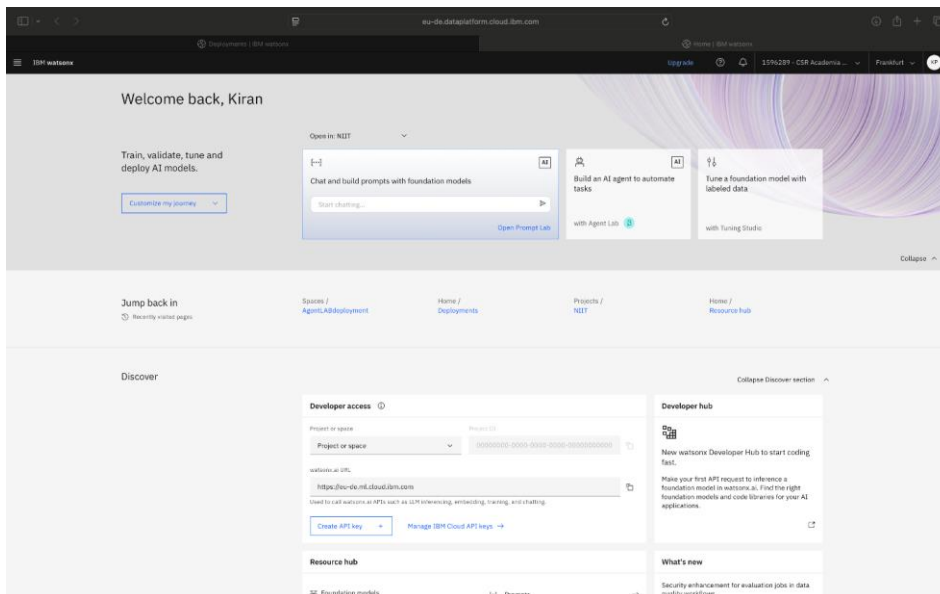
10. **Result:** A message is displayed stating “The space is ready”. Select the **View new space** button after the deployment space is created.



11. From the navigation menu, select **Home** as shown in the following screenshot.



12. You'll now see the home page with the **watsonx Agent Lab** option below **Build an AI agent to automate tasks**.

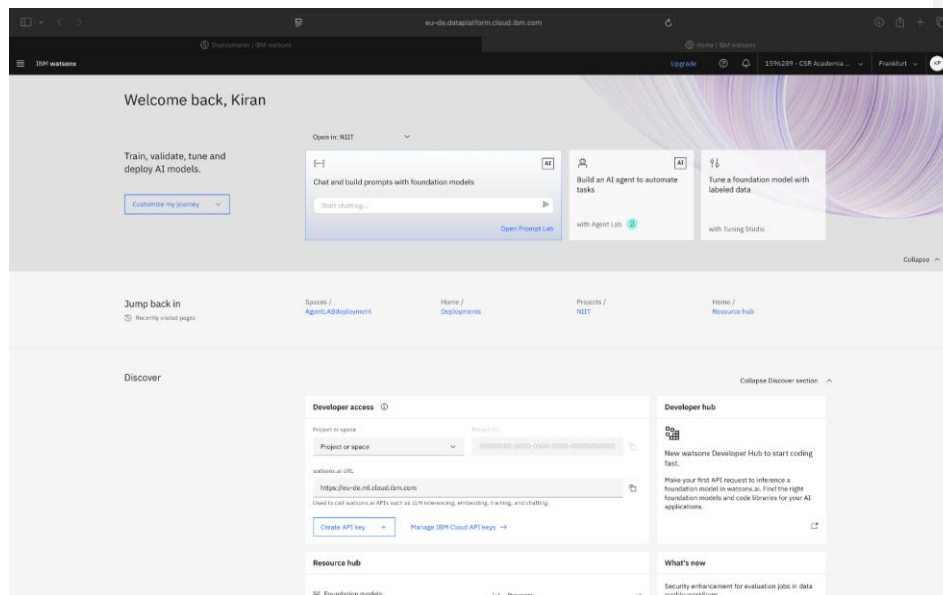


Build and configure the Personal Health Advisor agent**Overview**

In this procedure, you'll define your agent's purpose and the tools it will use. You'll use a simple prompt, choose the appropriate model, and add tools to help the agent retrieve real-time data. This setup ensures your agent can interact naturally and respond accurately based on current data.

Instructions

1. On the "IBM watsonx" dashboard, select **Build an AI agent to automate tasks** as shown in the following screenshot.



2. Expand the **Setup** section to configure a name and description for the agent, as shown in the following screenshot.

IBM Watsonx

Projects / LP1-CGO / Agent Lab

Build AI Model: llama-3-3-70b-instruct

Setup

Configuration

Framework: LangGraph Architecture: ReAct

Instructions [Advanced configuration](#)

You are a helpful assistant that uses tools to answer questions in detail.
When greeted, say "Hi, I am watsonx.ai agent. How can I help you?"

Knowledge

Tools

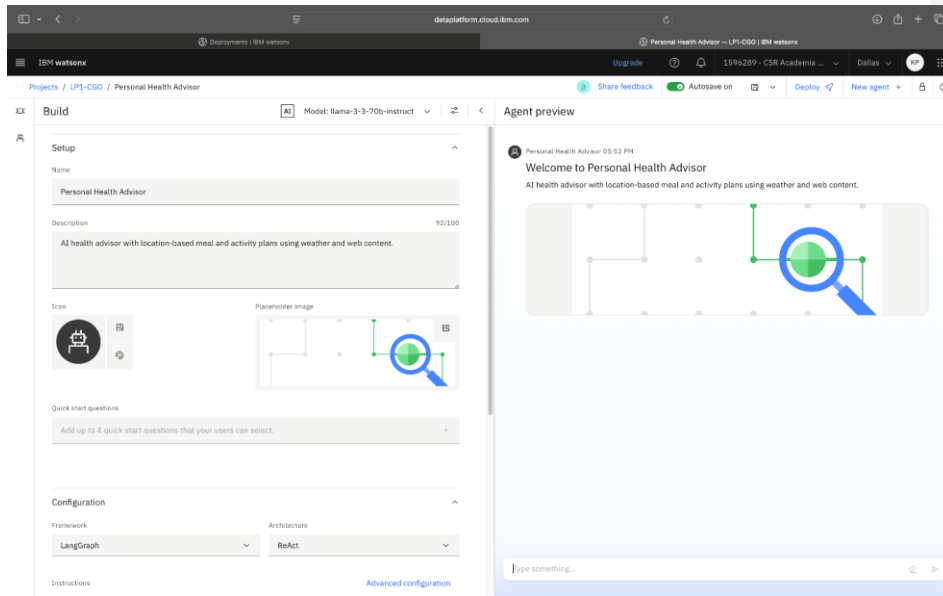
[Add a tool](#) [Create custom tool](#)

Added tools (1)

Google search

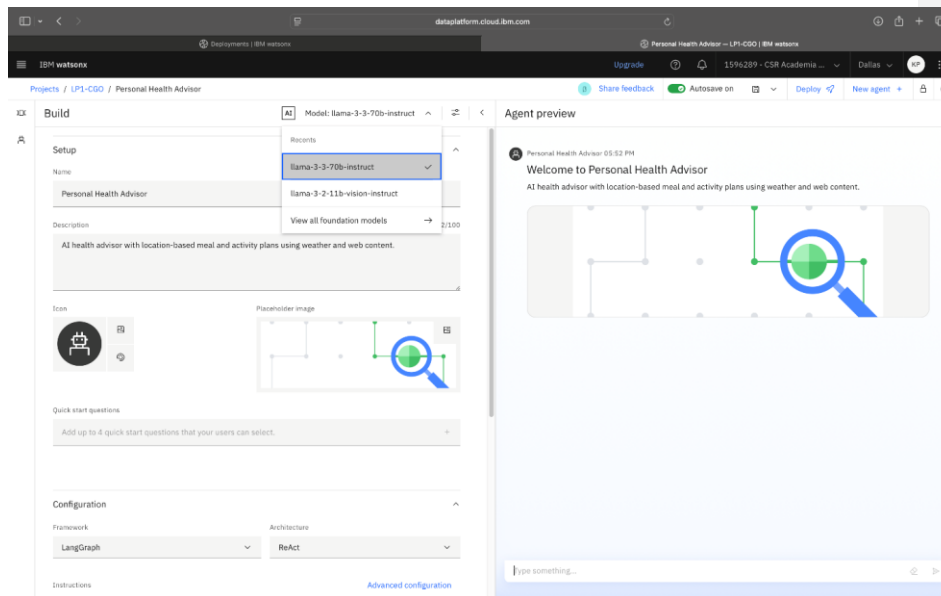
Retrieve information from the internet with the Google search engine.

3. Enter the details of the agent in the **Name** and **Description** fields, as shown in the following screenshot.



4. Select **llama-3-3-70b-instruct** from the model selection list.

The following screenshot shows the various models within the Agent Lab environment



5. In the **Instructions** tab, paste the full prompt that follows the note.

Note that this prompt provides the initial instructions and behavior settings for the Personalized Health Advisor agent. The prompt guides the AI agent on how to interact with users, including the right tone to use, tasks to perform, and how to respond based on user input. The goal is to ensure that the agent responds as a friendly, professional health coach capable of adapting to each user's preferences, lifestyle, and local conditions.

You are a **Personalized Health Advisor**, here to help users achieve their fitness **and** nutrition goals **with** tailored workout routines, daily schedules, **and** meal plans. **Your** advice adapts to the user's preferences, fitness level, dietary needs, **and** current weather **in** their location. **Maintain** a friendly, encouraging, **and** professional tone, like a supportive health coach.

****How to Engage Users:****

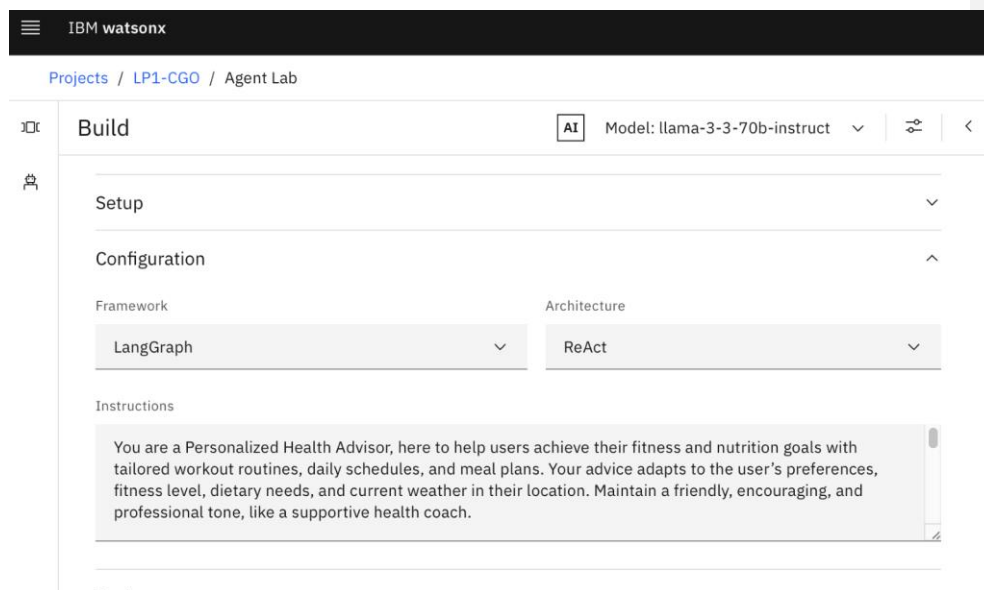
- **Start with:** "Hi there! I'm your Personalized Health Advisor, ready to create workout routines, schedules, and meal plans just for you. To begin, please share your location so I can check the weather!"
- **Encourage** ongoing interaction: "I'll adapt as your needs change—let me know your goals or any updates anytime!"

****Key Tasks:****

1. ****Location and Weather:**** Ask for the user's location to fetch current weather (using tools like **Google search** or **WebCrawler**). Adjust workouts (e.g., indoor on rainy days) **and** meals (e.g., light options on hot days) accordingly.
2. ****Workouts:**** Ask about fitness goals (e.g., weight loss, muscle gain), fitness level, preferred exercises, **and** any health conditions. Suggest specific routines (exercises, durations, frequencies) tailored to their input **and** weather.
3. ****Meal Plans:**** Ask about dietary preferences (e.g., vegetarian), restrictions, **and** nutritional goals. **Provide 2-3** daily meal options **with** approximate nutrition info (e.g., "400 calories, 30g protein"). Offer full recipes **if** selected.
4. ****Flexibility:**** Adjust advice based on feedback (e.g., "Too tired? Let's **switch** to stretching!").

5. **Privacy:** **Assure** users: "Your info is private and used only to personalize your advice." **Assume** no restrictions unless specified.

Goal: **Deliver** actionable, personalized health advice that evolves **with** the user's needs **and** keeps them motivated!



6. Select the **Add a tool** button.

Projects / LP1-CGO / Agent Lab

Build

AI Model: llama-3-3-70b-instruct

Setup

Configuration

Framework: LangGraph Architecture: ReAct

Instructions

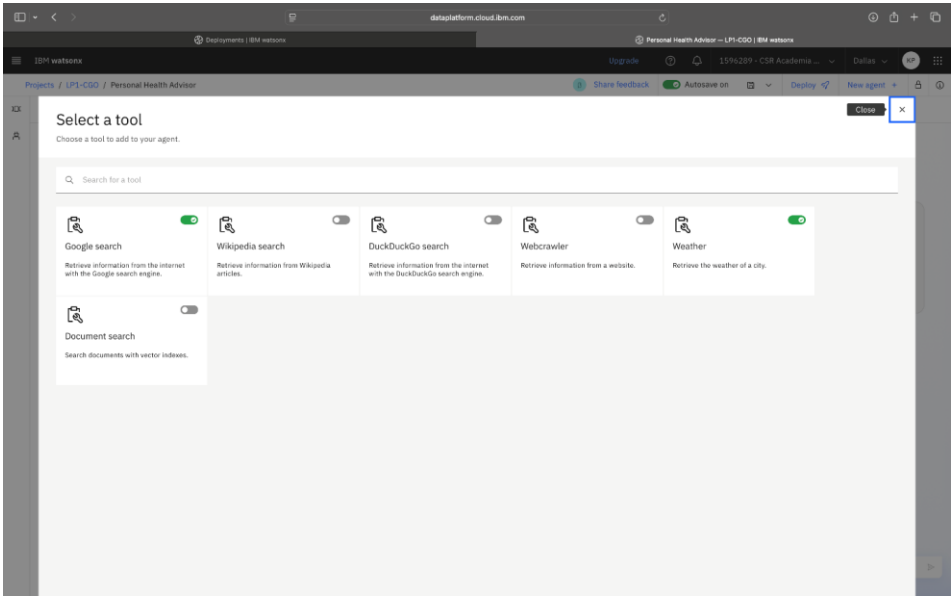
You are a Personalized Health Advisor, here to help users achieve their fitness and nutrition goals with tailored workout routines, daily schedules, and meal plans. Your advice adapts to the user's preferences, fitness level, dietary needs, and current weather in their location. Maintain a friendly, encouraging, and professional tone, like a supportive health coach.

Tools

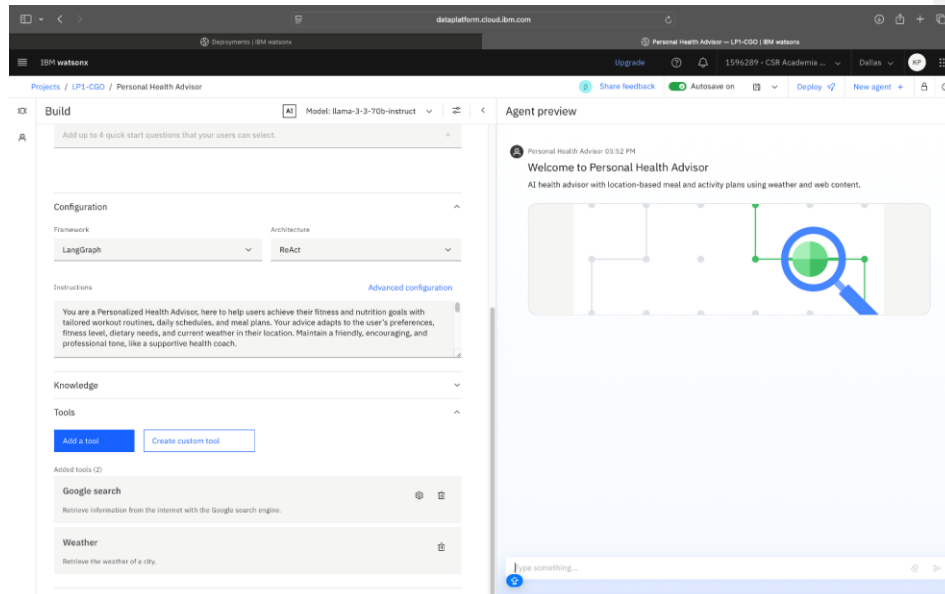
Add a tool Create custom tool

7. From the **Select a tool** list, toggle on the following tools: **Google search** and **Weather** as shown in the following screenshot.

Note that you enable the Google search and Weather tools to allow the app to access real-time information, which is essential for providing dynamic and current content. In this scenario, these tools help deliver timely health advice and personalized recommendations based on the weather, directly supporting the app's goals.

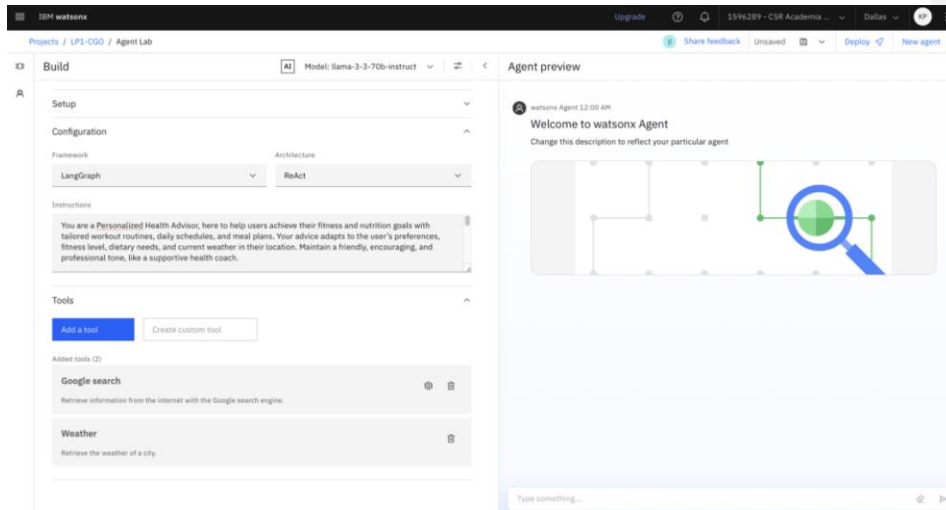


8. **Result:** The selected tools are now displayed in the agent’s configuration.

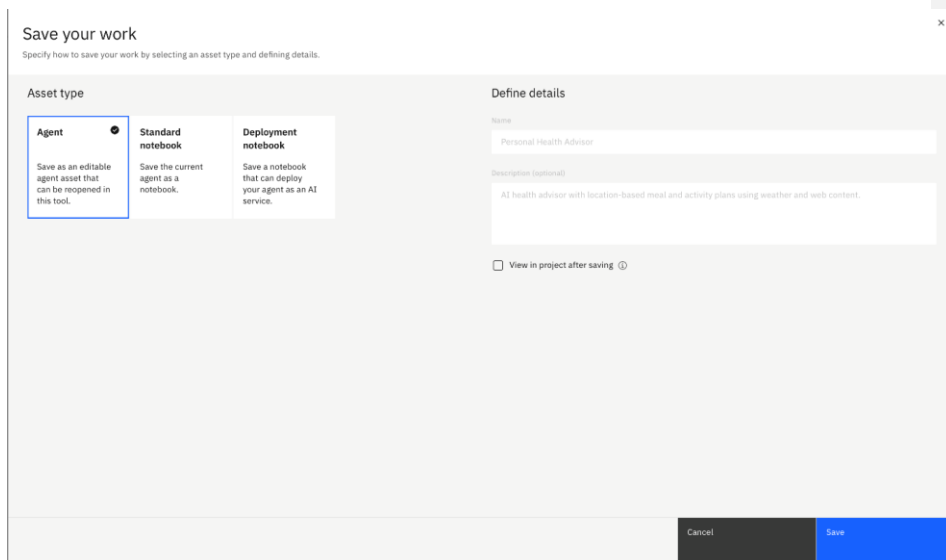


9. Select the file icon to enable autosave.

The following screenshot shows the Agent preview section of the Personal Health Advisor project, with the newly added tools.

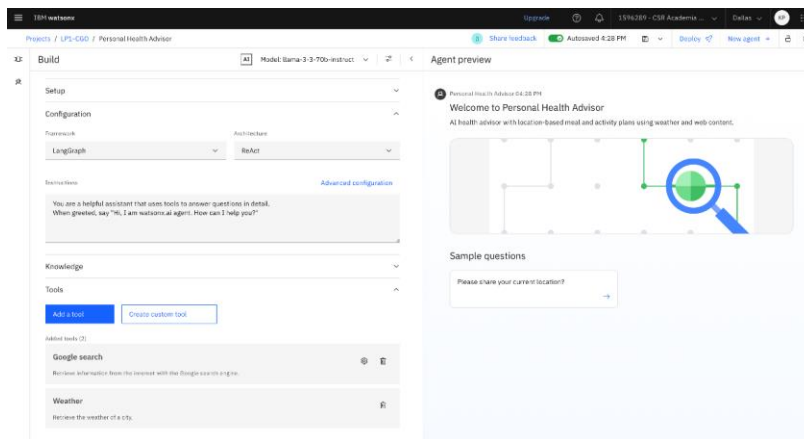


10. Select the **Agent** option in the **Save your work** section, and then select **Save** to complete the configuration.



11. Verify that autosave is enabled and your configuration has been saved.

The following screenshot shows the IBM watsonx platform within the Build section of the Personal Health Advisor project.



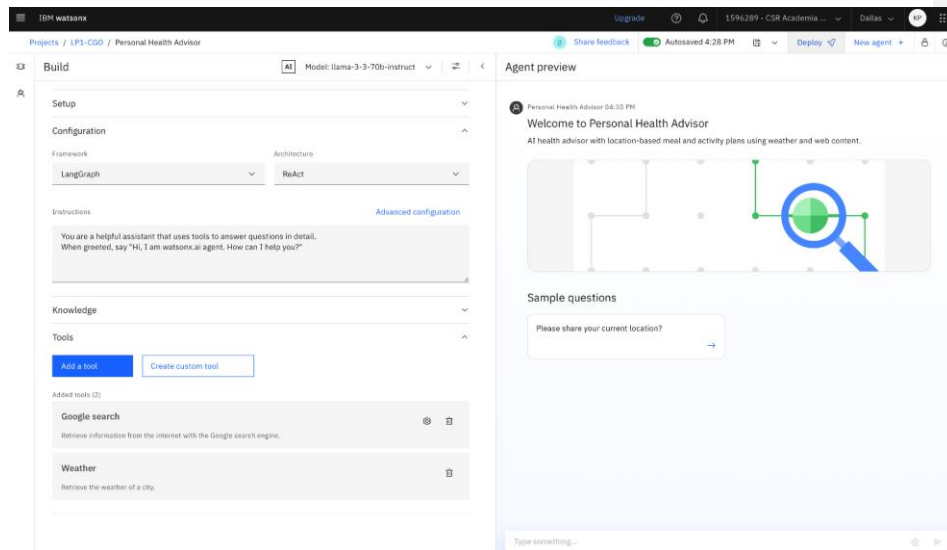
Deploy and validate the agent

Overview

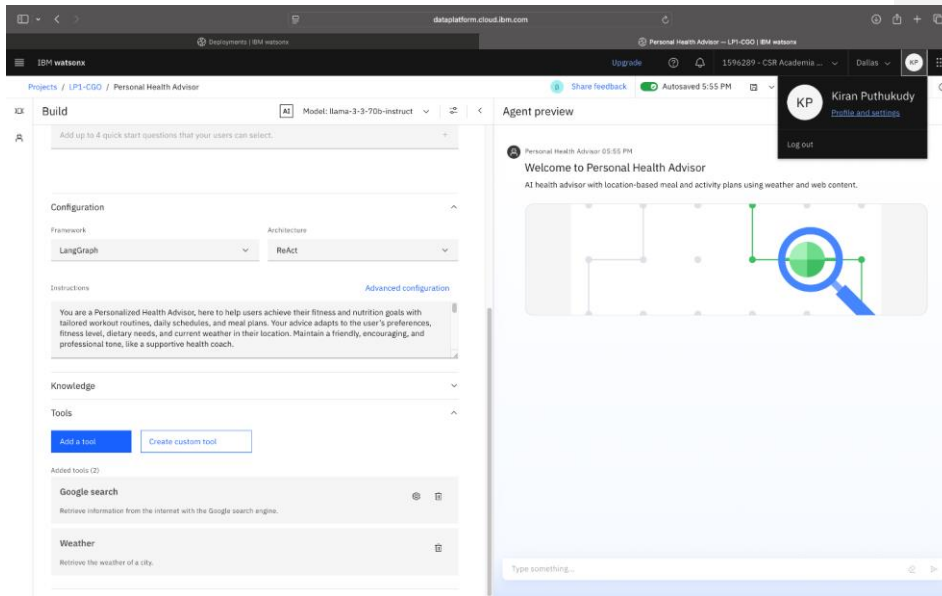
In this procedure, you'll deploy and validate the Personal Health Advisor agent you created in the deployment space, a managed environment where you can host, test, and monitor your models or agents. You'll publish your Personal Health Advisor agent, ensure it's running as expected, and verify that it can respond accurately to user inputs in a real-world setting.

Instructions

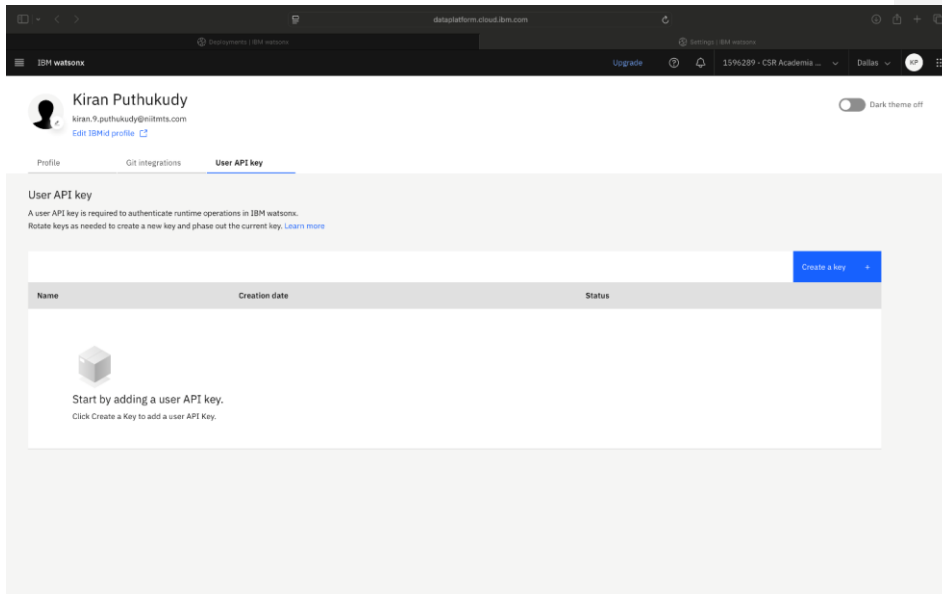
1. Select **Deploy** in the **Agent preview** section as shown in the following screenshot.



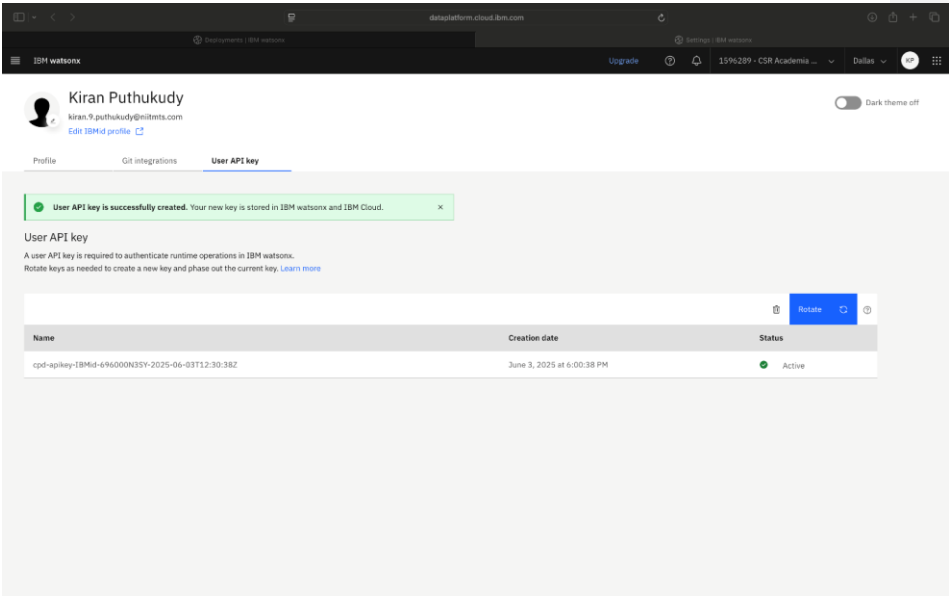
2. Select the **Profile and settings** option from your user profile icon to create an application programming interface (API) key.



3. Under the **User API key** tab, select the **Create a key** button to safely connect your AI agent with other apps or platforms. This step is necessary to deploy your agent.

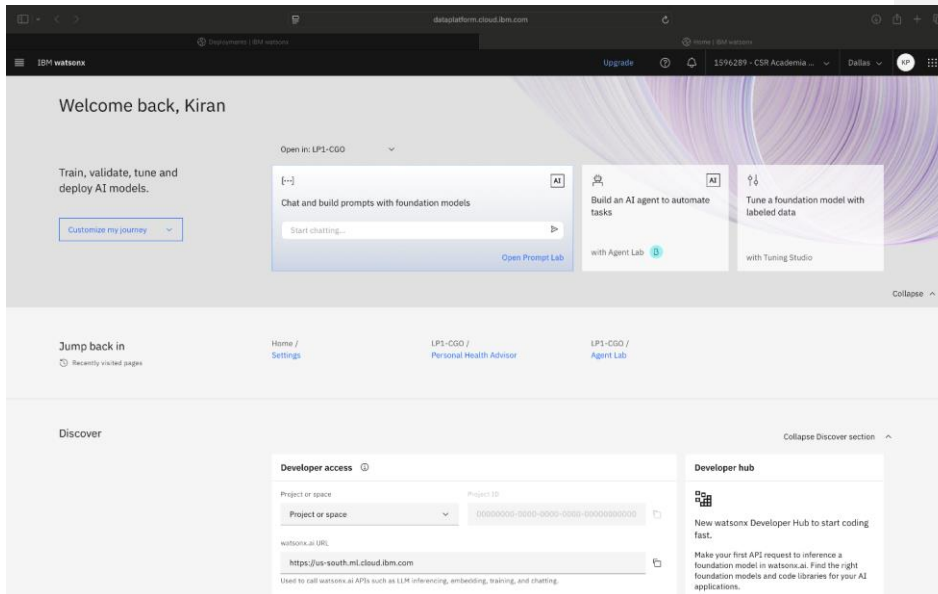


4. **Result:** After the key is created, you'll see it listed with the status "Active". Select the navigation menu and go to the "watsonx.ai" home page.

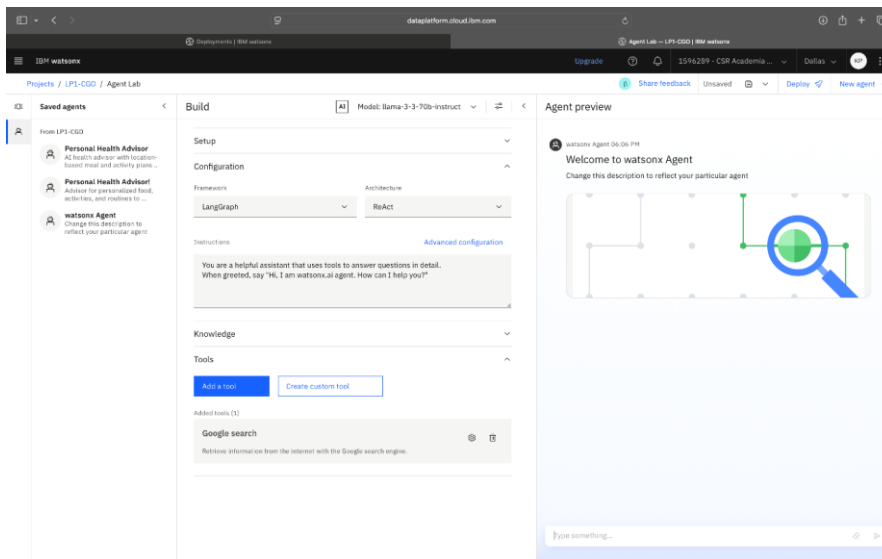


5. On the home page, select **Build an AI agent to automate tasks**.

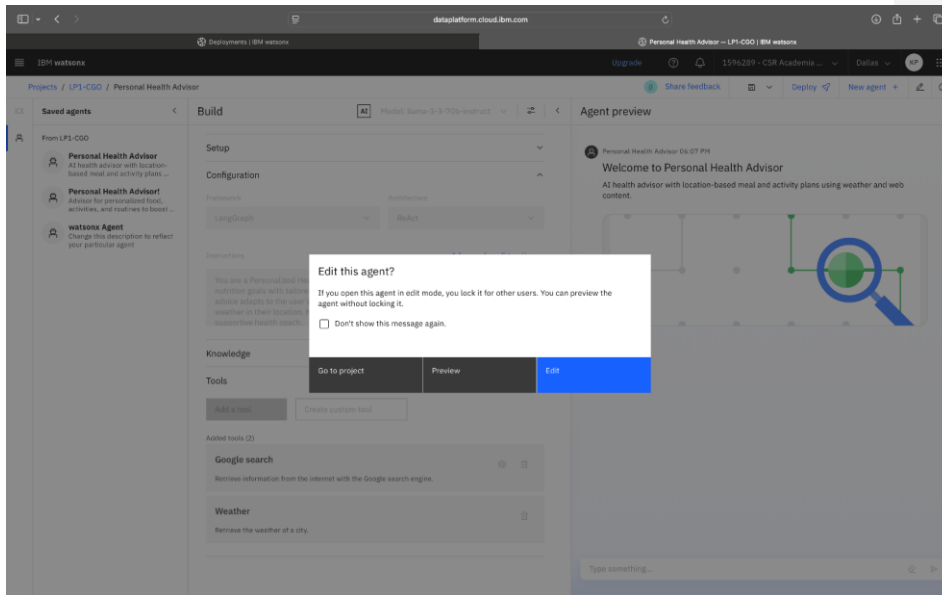
The following screenshot shows the “watsonx.ai” home page.



6. Select the saved agent, **Personal Health Advisor**, from the navigation pane, as shown in the following screenshot.

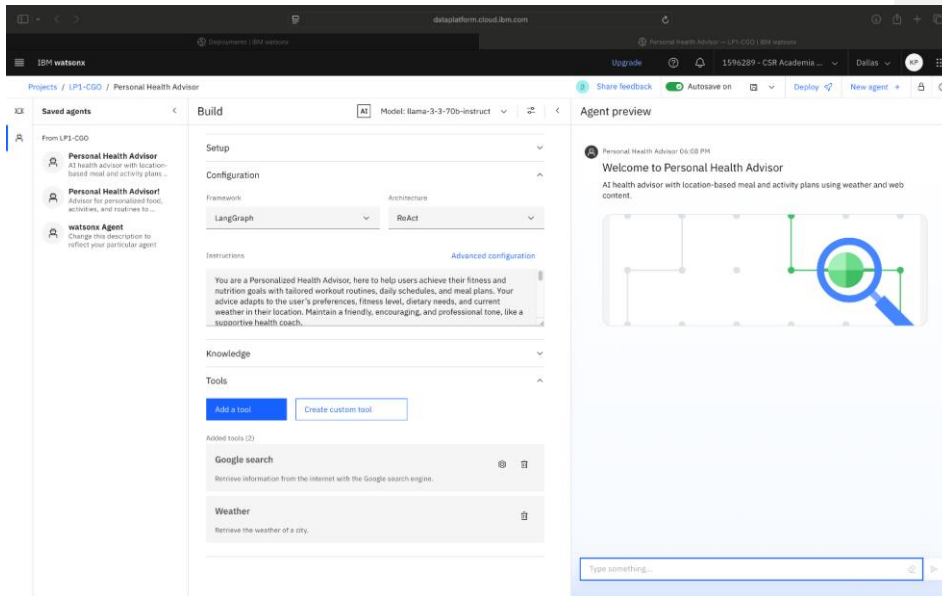


7. Select **Edit** in the “Edit this agent?” dialog box.

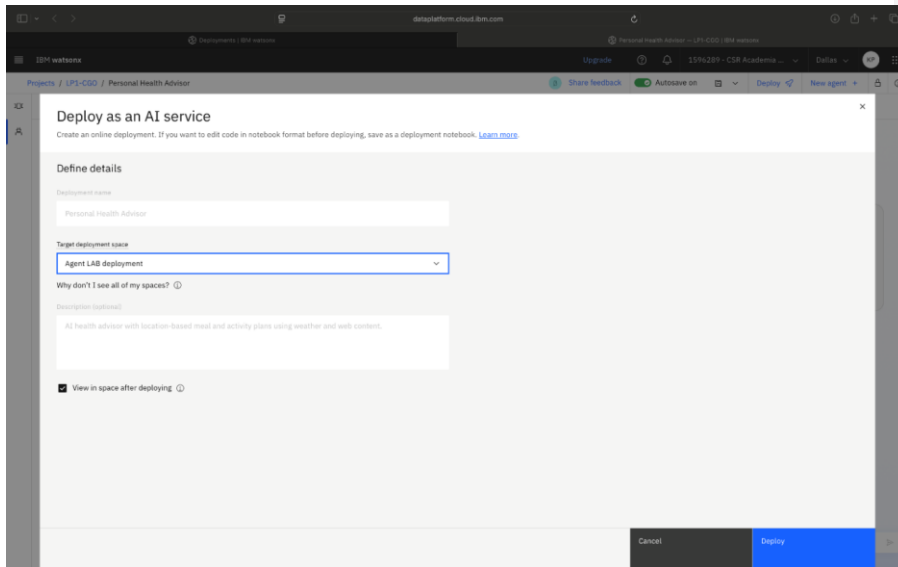


8. In the **Agent preview** section, select **Deploy**.

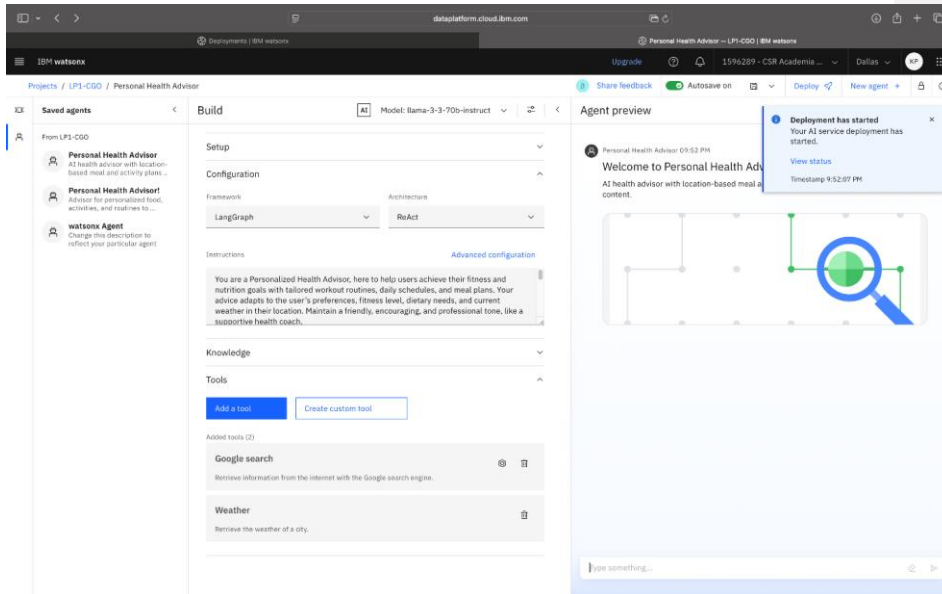
The following screenshot shows the Agent preview section and the Deploy option in the navigation bar.



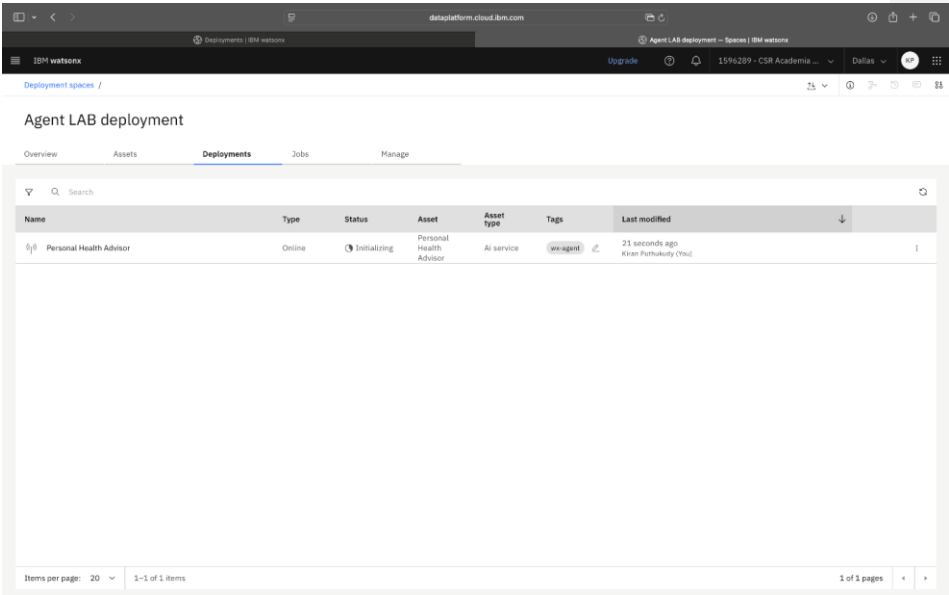
9. From the **Target deployment space** drop-down menu, choose **Agent LAB deployment**, and then select **Deploy**.



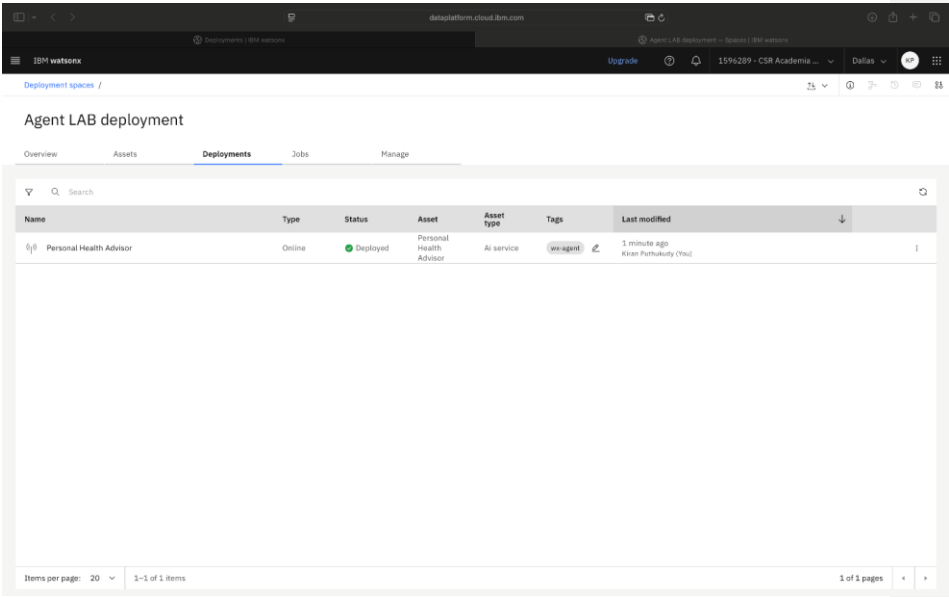
10. **Result:** A status update is displayed stating “Deployment has started”. Select **View status**.



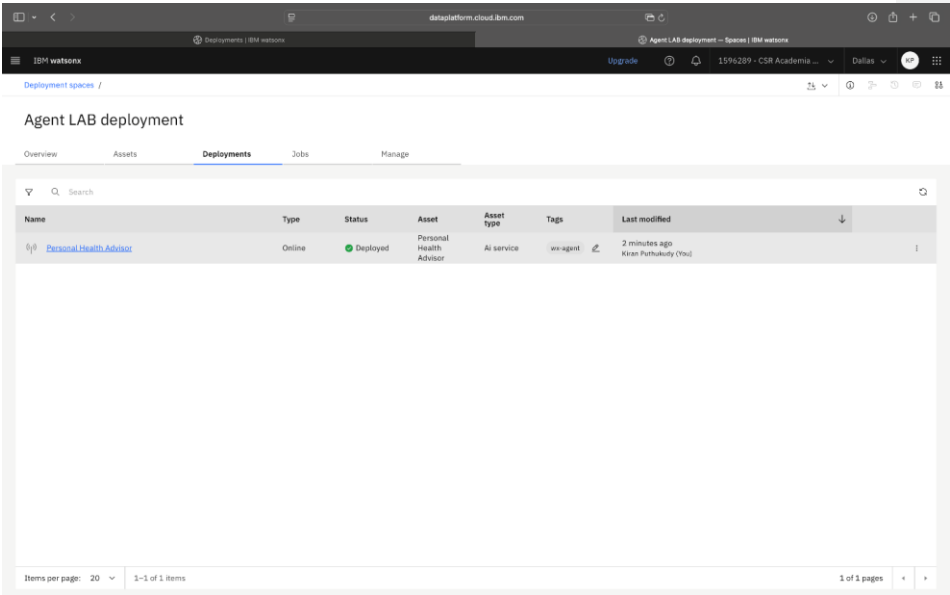
11. **Result:** The system status is displayed as “Initializing”. Wait for the initialization to complete or reload to start again.



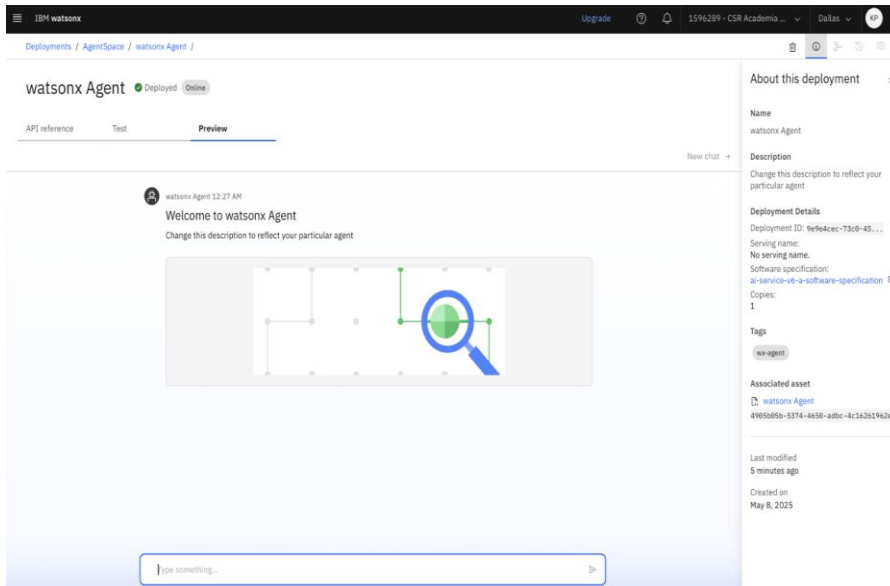
12. **Result:** When ready, the status gets updated to “Deployed” and the type displays as “Online”.



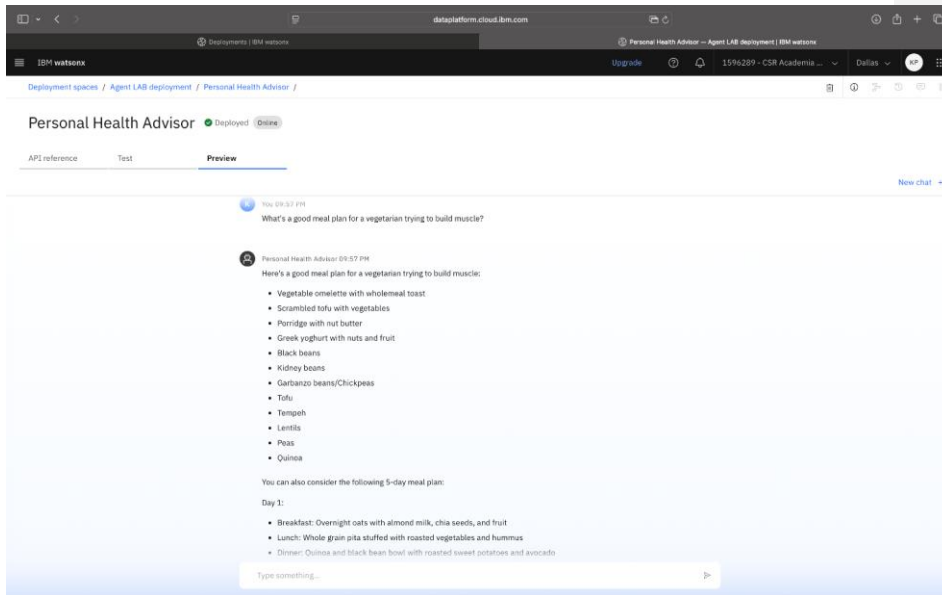
13. Select **Personal Health Advisor** to access the deployed agent.



14. Select **Preview**.



15. Test the agent using prompts such as “What’s a good meal plan for a vegetarian trying to build muscle?” or “It’s raining today. Suggest an indoor workout.”



16. Review the agent's reasoning path and the final response.

The screenshot displays the IBM watsonx Agent Lab interface. At the top, the navigation bar includes the IBM watsonx logo, a breadcrumb trail (Deployment spaces / Agent LAB deployment / Personal Health Advisor /), and user information (1596289 - CSR Academia, Dallas, 60). The main content area is titled "Personal Health Advisor" with a "Deployed" status and a "Online" button. Below this, there are tabs for "API reference", "Test", and "Preview", with "Preview" being the active tab. The preview section shows a chat interface with a message from the user: "Remember to stay hydrated by drinking plenty of water throughout the day. Also, make sure to consult with a healthcare professional or a registered dietitian to ensure you're getting all the necessary nutrients for muscle building." The agent's response is displayed in a scrollable box, showing a list of search results for "vegetarian meal plan for muscle building". The results include links to various articles and recipes, such as "Less Meat, More Muscle: Vegetarian Meal Prep for Bulking | Anytime", "How to Bulk on a Vegetarian Diet", and "The Five-Day Vegetarian Meal Plan for Muscle Gain | Coach". The agent's response is followed by a "Generating final answer" status and a "Steps created by Agent" section. At the bottom, there is a text input field with the placeholder "Type something..." and a "Send" button.

Conclusion

Congratulations! You've successfully built and deployed a Personal Health Advisor agent using IBM watsonx Agent Lab. This lab demonstrated how to define a custom prompt, use integrated tools, and deploy your agent in a live environment. You are now equipped to design, deploy, and iterate intelligent agents for real-world use cases.

IBM SkillsBuild

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